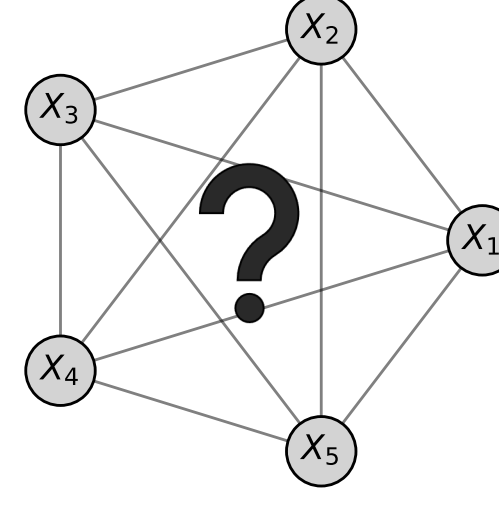




## Problem statement

- Learn causal relations between variables as a directed, acyclic graph (DAG) from observational and interventional data
- Assumptions: interventions are sparse (only one variable at a time), soft (distribution over values), perfect (new distribution independent of original parents), and available for all variables.
- Identifiability has been theoretically studied under this setting, but in practice, computational methods yet struggle to recover large graphs
- Continuous-optimization methods are promising since they are efficient by leveraging expressiveness of neural networks, but acyclicity needs to be ensured by constrained optimization methods which are slow and sensitive to hyperparameters, or regularizers without guarantess



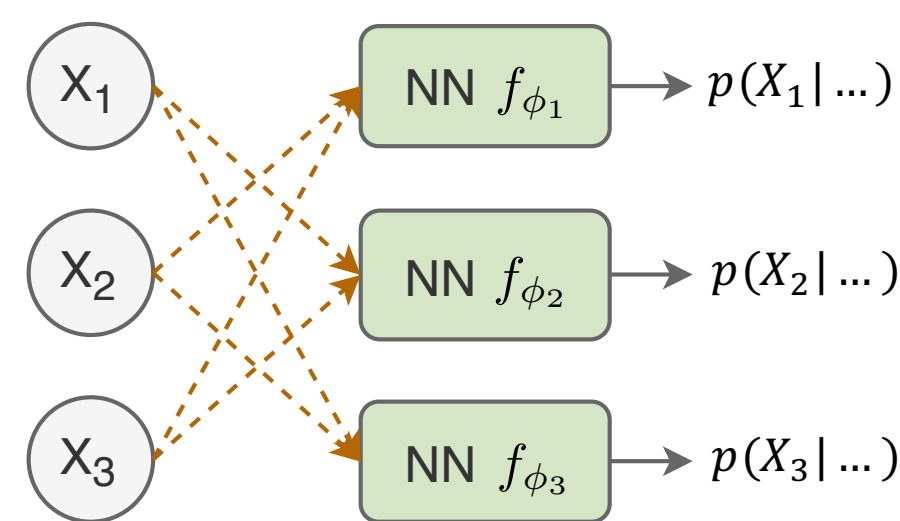
## ENCO – Efficient Neural Causal Discovery

- Optimize likelihoods of edges based on how well graphs generalize from observational to interventional data
- Split parameters into two groups: edge existence  $\gamma$  and orientation  $\theta$  for better control on gradient updates
- Probability of edges:  $p(X_i \rightarrow X_j) = \sigma(\gamma_{ij}) \cdot \sigma(\theta_{ij})$
- Fit observational distributions by neural networks, and evaluate different graphs on likelihood-based score function with interventional data:

$$\tilde{\mathcal{L}} = \mathbb{E}_{\hat{I} \sim p_I(I)} \mathbb{E}_{\hat{p}_I(\mathbf{X})} \mathbb{E}_{p_{\gamma, \theta}(C)} \left[ \sum_{i=1}^N \mathcal{L}_C(X_i) \right] + \lambda_{\text{sparse}} \sum_{i=1}^N \sum_{j=1}^N \sigma(\gamma_{ij}) \cdot \sigma(\theta_{ij})$$

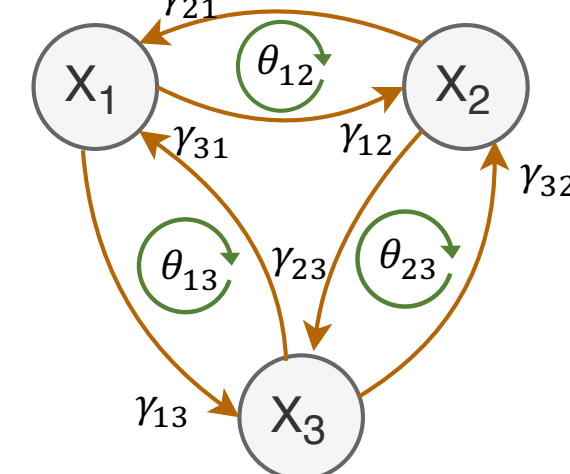
- No regularizer or constraint for acyclicity needed!

### Distribution fitting

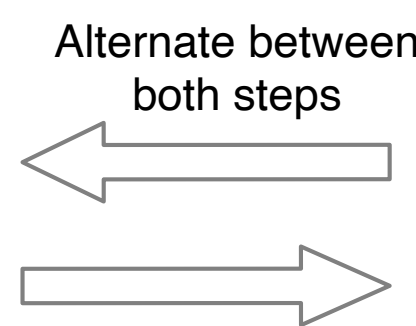


→ Fit NNs to conditional, observational distributions

### Graph fitting



→ Learn edge and orientation parameters based on fitted distributions

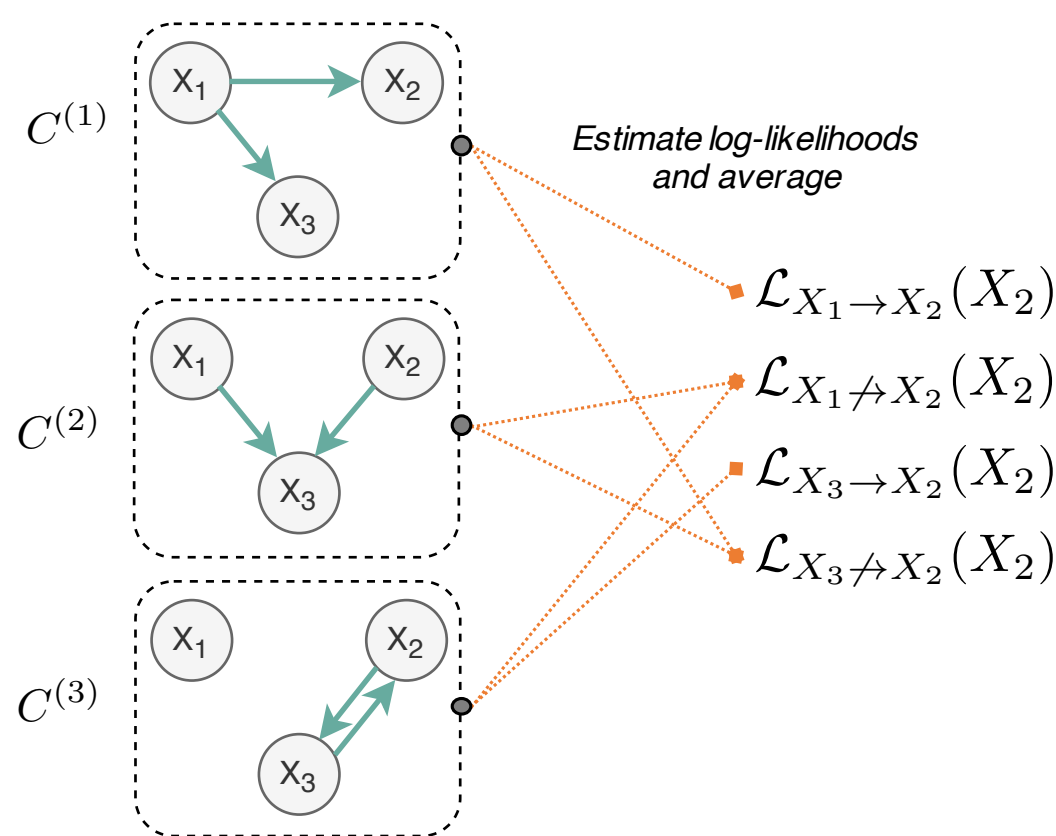
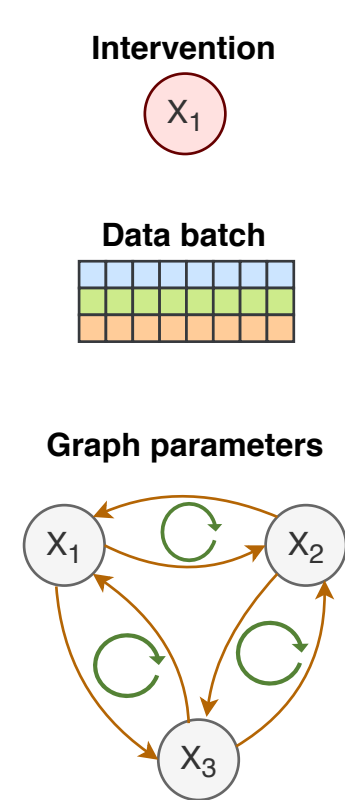


## Graph optimization

- Unbiased, low-variance gradient estimator for  $\gamma$  and  $\theta$  via REINFORCE and Monte-Carlo sampling
- Intuition: sample an interventional data batch and  $K$  graphs, and check for each edge whether on average, its existence improved likelihood estimate of the child

$$\frac{\partial}{\partial \gamma_{ij}} \tilde{\mathcal{L}} = \sigma'(\gamma_{ij}) \cdot \sigma(\theta_{ij}) \cdot \mathbb{E}_{\mathbf{X}, C \sim ij} [\mathcal{L}_{X_i \rightarrow X_j}(X_j) - \mathcal{L}_{X_i \not\rightarrow X_j}(X_j) + \lambda_{\text{sparse}}]$$

- Orientation parameters are only updated on interventions on adjacent nodes since this is the only reliable gradient information to distinguish directions



$$\begin{aligned} \frac{\partial}{\partial \gamma_{12}} \tilde{\mathcal{L}} &= \dots \\ \frac{\partial}{\partial \gamma_{32}} \tilde{\mathcal{L}} &= \dots \\ \frac{\partial}{\partial \theta_{12}} \tilde{\mathcal{L}} &= \dots \end{aligned}$$

## Latent confounders

- Latent confounders between two observable variables that do not have a direct causal relation between each other, cause a unique pattern in the graph parameters
- An edge between the two variables is disadvantageous on interventional data but beneficial when intervening on any other variable
- Phenomenon can be detected by recording observational and interventional gradients on  $\gamma$  separately, and combine in a score function:

$$\text{lc}(X_i, X_j) = \sigma(\gamma_{ij}^{(O)}) \cdot \sigma(\gamma_{ji}^{(O)}) \cdot (1 - \sigma(\gamma_{ij}^{(I)})) \cdot (1 - \sigma(\gamma_{ji}^{(I)}))$$

## Convergence

- Under mild conditions, we can guarantee that ENCO converges to the correct, directed acyclic causal graph despite not explicitly constraining search space to acyclic graphs
- Equivalent to showing that there is no local optimum for graph parameters
- Central condition: the expected improvement of the likelihood estimate by adding a correct edge must be greater than the sparsity regularizer  $\lambda_{\text{sparse}}$

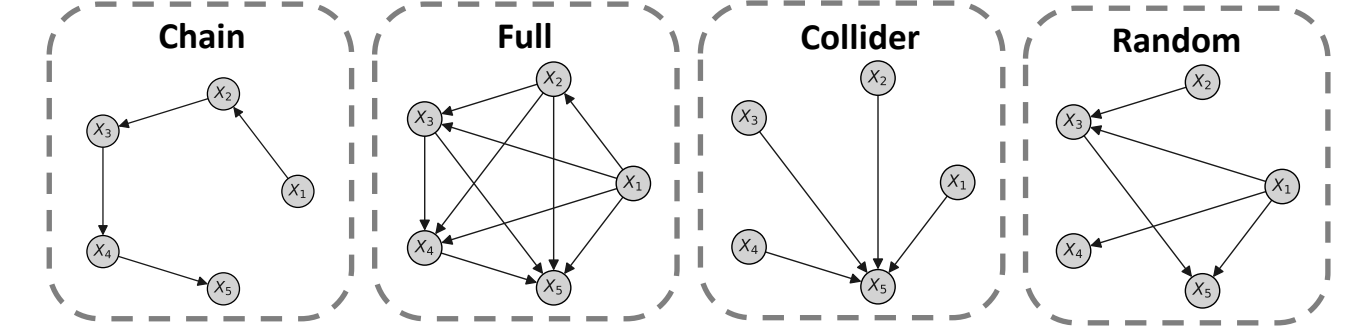
$$\min_{\text{pa} \subseteq \text{gpa}_i(X_j)} \mathbb{E}_{\hat{I} \sim p_I(I)} \mathbb{E}_{\hat{p}_I(\mathbf{X})} [\log p(X_j | \text{pa}, X_i) - \log p(X_j | \text{pa})] > \lambda_{\text{sparse}}$$

- Tradeoff: low values of  $\lambda_{\text{sparse}}$  might require longer training times

## Experiments

### Synthetic graphs

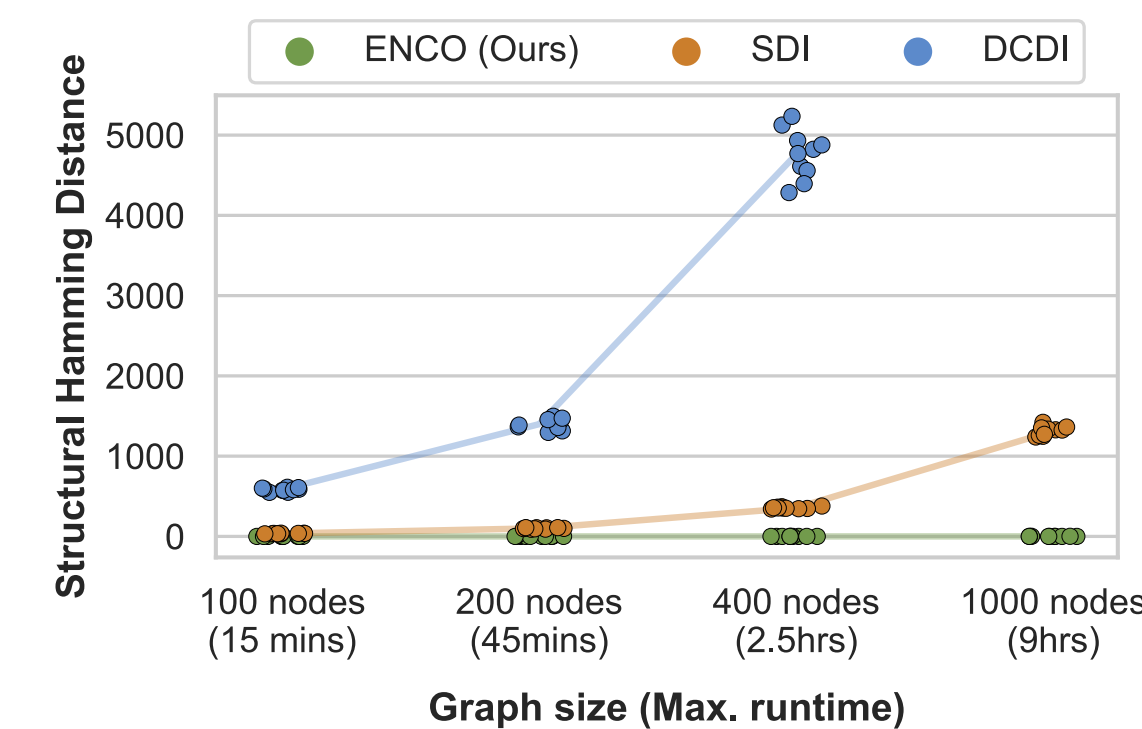
- Synthetically generated graphs with 25 categorical variables of different structures
- ENCO recovers 5 out of 6 graph structures without errors



| Graph type                       | bidiag                   | chain                    | collider                 | full                     | jungle                   | random                   |
|----------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| GIES [Hauser and Bühlmann, 2012] | 47.4 ( $\pm 5.2$ )       | 22.3 ( $\pm 3.5$ )       | 13.3 ( $\pm 3.0$ )       | 152.7 ( $\pm 12.0$ )     | 53.9 ( $\pm 8.9$ )       | 86.1 ( $\pm 12.0$ )      |
| IGSP [Wang et al., 2017]         | 33.0 ( $\pm 4.2$ )       | 12.0 ( $\pm 1.9$ )       | 23.4 ( $\pm 2.2$ )       | 264.6 ( $\pm 7.4$ )      | 38.6 ( $\pm 5.7$ )       | 76.3 ( $\pm 7.7$ )       |
| SDI [Ke et al., 2019]            | 2.1 ( $\pm 1.5$ )        | 0.8 ( $\pm 0.9$ )        | 14.7 ( $\pm 4.0$ )       | 121.6 ( $\pm 18.4$ )     | 1.8 ( $\pm 1.6$ )        | 1.8 ( $\pm 1.9$ )        |
| DCDI [Brouillard et al., 2020]   | 3.7 ( $\pm 1.5$ )        | 4.0 ( $\pm 1.3$ )        | <b>0.0</b> ( $\pm 0.0$ ) | 2.8 ( $\pm 2.1$ )        | 1.2 ( $\pm 1.5$ )        | 2.2 ( $\pm 1.5$ )        |
| ENCO (Ours)                      | <b>0.0</b> ( $\pm 0.0$ ) | <b>0.0</b> ( $\pm 0.0$ ) | <b>0.0</b> ( $\pm 0.0$ ) | <b>0.3</b> ( $\pm 0.9$ ) | <b>0.0</b> ( $\pm 0.0$ ) | <b>0.0</b> ( $\pm 0.0$ ) |

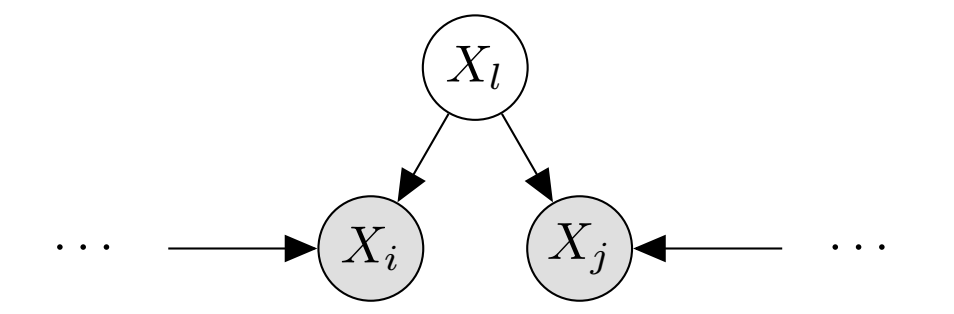
### Scalability

- Testing ENCO and strongest baselines on scalability to large, synthetic graphs of 100 to 1000 variables
- With same NN and hardware setting, ENCO is much more efficient making only two errors overall in 10 graphs of 1000 variables



### Latent confounders

- Synthetic graphs with 25 variables plus 5 latent confounders on arbitrary node pairs except parent-child relations
- ENCO detects most confounders while missed confounders don't effect the SHD



| Metrics              | ENCO                   |
|----------------------|------------------------|
| SHD                  | 0.0 ( $\pm 0.0$ )      |
| Confounder recall    | 96.8% ( $\pm 9.5\%$ )  |
| Confounder precision | 100.0% ( $\pm 0.0\%$ ) |

## BnLearn Repository

- Real-world inspired causal graphs from the BnLearn Repository [Scutari, 2010] scaling from 5 to 441 variables
- More challenging because of strongly peaked marginal distribution and deterministic variables
- ENCO achieves perfect reconstruction for most graphs, and even works well on *diabetes* with many deterministic variables

| Dataset               | cancer (5 nodes) | asia (8 nodes) | sachs (11 nodes) | child (20 nodes) | alarm (37 nodes) | diabetes (413 nodes) | pigs (441 nodes) |
|-----------------------|------------------|----------------|------------------|------------------|------------------|----------------------|------------------|
| SDI [Ke et al., 2019] | 3.0              | 4.0            | 7.0              | 11.8             | 24.6             | 422.4                | 18.0             |
| ENCO (Ours)           | <b>0.0</b>       | <b>0.0</b>     | <b>0.0</b>       | <b>0.0</b>       | <b>1.0</b>       | <b>2.0</b>           | <b>0.0</b>       |

## Takeaways

- Splitting graph parameters into edge existence and orientation for greater control over gradients without a need for acyclicity constraints
- Efficient graph optimization using low-variance gradient estimators by evaluating a set of graphs on their generalization to interventions
- Convergence guarantees can be given under mild conditions with respect to likelihood improvements per edge
- ENCO reliably and efficiently recovers causal graphs of various structures with up to 1000 variables, including latent confounders

Check out our paper and code for details!